

LABORATORY EQUIPMENT BOOKING & MAINTENANCE SYSTEM

DATABASE MANAGEMENT SYSTEM PROJECT REPORT

STUDENT NAME	Mallavarapu Suchith
ROLL NUMBER	25WU0102144
PROGRAM	B.Tech Artificial Intelligence and Machine Learning
COURSE	Database Management Systems (DBMS)
FACULTY / GUIDE	Dr. Kiran Mayee Adavala
ACADEMIC YEAR	2026–27
DATABASE	lab_equipment

Laboratory Equipment Booking and Maintenance System

Project Report

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1. ABSTRACT

The Laboratory Equipment Booking and Maintenance System is a relational database management system designed to centralize laboratory resource management. It stores information about laboratories, equipment categories, equipment, users, time slots, bookings, usage logs, consumables, calibrations, breakdowns, maintenance jobs and vendors.

The implementation uses MySQL 8.0.46 with primary keys, foreign keys, UNIQUE, NOT NULL, DEFAULT and CHECK constraints. Database triggers validate important booking rules, while views and SQL queries provide operational reports. A web-based user interface is used as the front end for displaying and demonstrating database records and CRUD operations.

The project demonstrates core DBMS concepts including entity–relationship modeling, relational schema design, normalization up to Third Normal Form, integrity constraints, joins, aggregate functions, nested queries, views and database-level validation.

2. INTRODUCTION AND PROBLEM STATEMENT

2.1 Introduction

Laboratories contain expensive and shared equipment that must be scheduled carefully. Students, faculty and technicians may need the same equipment at different times. In addition to bookings, equipment can require calibration, may develop faults and may need maintenance by external vendors. Consumables also need to be monitored so that laboratory work is not interrupted.

2.2 Problem Statement

Manual or disconnected records can result in double booking, unclear equipment status, missing usage history, delayed maintenance and poor visibility of calibration and stock levels. The proposed system addresses these issues through a centralized relational database.

2.3 Users

USER	RESPONSIBILITIES
Student	View available equipment, create bookings, view booking status and record laboratory usage.
Faculty	Book equipment, supervise laboratory work and view equipment and booking reports.
Technician	View breakdowns, manage maintenance jobs and update equipment maintenance status.
Administrator	Maintain master records, users, laboratories, equipment, vendors and database configuration.

2.4 Functional Requirements

- Laboratory management
- Equipment-category management
- Equipment management
- User management
- Time-slot management
- Equipment booking and booking-status management
- Usage-log recording
- Consumable stock management
- Calibration tracking
- Breakdown reporting
- Maintenance-job management
- Vendor management
- Reports using SQL queries and views
- CRUD operations through the web-based UI
- Validation using foreign keys, CHECK constraints and triggers

3. OBJECTIVES, USERS AND SCOPE

3.1 Objectives

- Design a relational database for laboratory equipment booking and maintenance.
- Normalize the database to Third Normal Form (3NF).
- Enforce data integrity using PK, FK, UNIQUE, NOT NULL, DEFAULT and CHECK constraints.
- Maintain equipment, laboratories, users and time-slot information.
- Support equipment bookings and usage tracking.
- Track consumables, calibration due dates and breakdowns.
- Record maintenance jobs and vendor details.
- Demonstrate joins, aggregate queries, nested queries and views.
- Connect a web-based UI to the database and demonstrate CRUD operations.

3.2 Scope

The system covers the operational database required for a laboratory equipment management prototype. The implemented database contains 12 principal base tables. Four views are provided for reusable reporting. Authentication, enterprise notification services, mobile applications and automated external integrations are treated as future enhancements rather than required components of the current implementation.

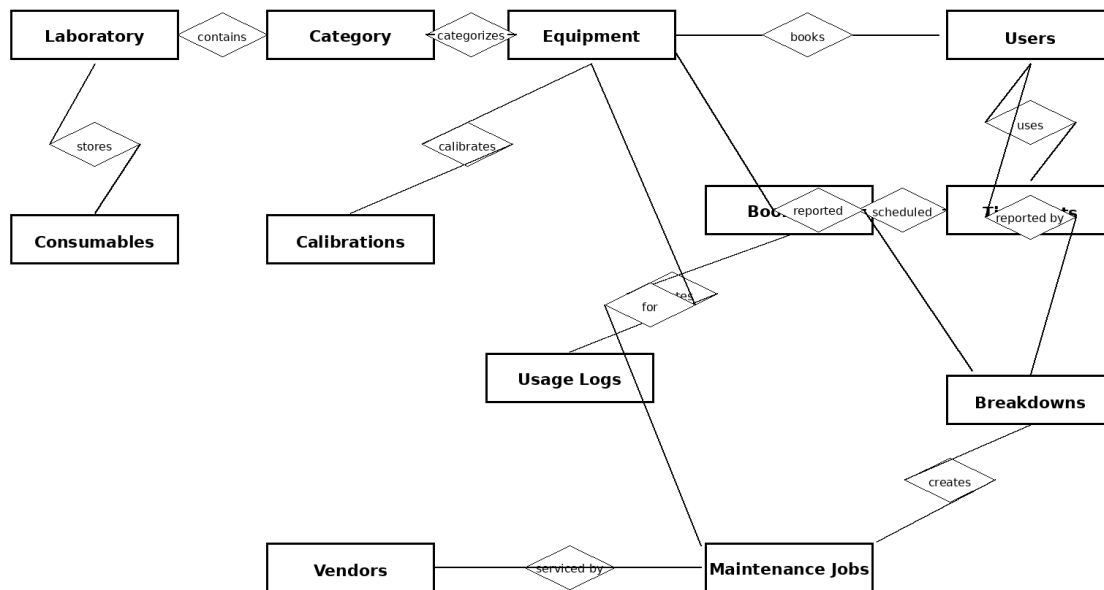
4. SOFTWARE AND HARDWARE REQUIREMENTS

CATEGORY	REQUIREMENT	PURPOSE
Operating System	Windows 10/11 or compatible OS	Development and execution
Language / Database	SQL / MySQL 8.0.46	Database implementation
Database Tool	MySQL Workbench / MySQL CLI	Administration and verification
UI	Web-based user interface	CRUD and report presentation
Development	Visual Studio Code / project source files	Source-code development
Version Control	Git / GitHub	Project submission
Processor	Dual-core or better	Application execution
RAM	4 GB minimum; 8 GB recommended	Development
Storage	At least 1 GB free	Source code, tools and report
Browser	Modern web browser	Accessing the UI

The project database is implemented in MySQL 8.0.46. MySQL CLI is used for SQL execution and MySQL Workbench can be used for schema visualization and verification.

5. ER DIAGRAM

The conceptual model represents the main entities and relationships in the Laboratory Equipment Booking and Maintenance System. The implemented relational model contains 12 principal entities.



Chen notation: rectangles = entities; diamonds = relationships. Detailed attributes and PK/FK definitions are in the data dictionary.

Figure 1. Entity–Relationship diagram for the Laboratory Equipment Booking and Maintenance System.

5.1 Key Relationships and Cardinalities

- Laboratory 1:N Equipment
- Equipment_Category 1:N Equipment
- Laboratory 1:N Consumables
- Users 1:N Bookings
- Equipment 1:N Bookings
- Time_Slots 1:N Bookings
- Bookings 1:N Usage_Logs
- Equipment 1:N Calibrations
- Equipment 1:N Breakdowns
- Users 1:N Breakdowns
- Breakdowns 1:N Maintenance_Jobs
- Vendors 1:N Maintenance_Jobs
- Equipment 1:N Maintenance_Jobs

5.2 Business Rules

- Only active equipment should be bookable.
- An approved booking must not overlap another approved booking for the same equipment and date.
- A booking must reference an existing user, equipment and time slot.
- A maintenance job must reference a valid breakdown, vendor and equipment.
- Consumable quantities cannot be negative.

- Calibration next_due_date must be later than calibration_date.

6. RELATIONAL SCHEMA

The following relations correspond to the implemented 12 base tables. Primary keys identify each record and foreign keys connect related entities.

TABLE	RELATIONAL SCHEMA
Laboratory	LABORATORY(lab_id PK, lab_name, location)
Equipment_Category	EQUIPMENT_CATEGORY(category_id PK, category_name, description)
Equipment	EQUIPMENT(equipment_id PK, equipment_name, category_id FK, lab_id FK, status)
Users	USERS(user_id PK, name, email, role)
Time_Slots	TIME_SLOTS(slot_id PK, start_time, end_time)
Bookings	BOOKINGS(booking_id PK, user_id FK, equipment_id FK, slot_id FK, booking_date, status)
Usage_Logs	USAGE_LOGS(log_id PK, booking_id FK, equipment_id FK, user_id FK, actual_start, actual_end, purpose)
Consumables	CONSUMABLES(consumable_id PK, lab_id FK, name, quantity, minimum_stock)
Calibrations	CALIBRATIONS(calibration_id PK, equipment_id FK, calibration_date, next_due_date, status)
Breakdowns	BREAKDOWNS(breakdown_id PK, equipment_id FK, reported_by FK, reported_date, severity, description, status)
Vendors	VENDORS(vendor_id PK, vendor_name, contact, email)
Maintenance_Jobs	MAINTENANCE_JOBS(job_id PK, breakdown_id FK, vendor_id FK, equipment_id FK, start_date, completion_date, status, remarks)

Database objects in addition to the base tables

VIEW	PURPOSE
Equipment_Overview	Equipment joined with category and laboratory details.
Booking_Report	Bookings joined with user, equipment, laboratory and time-slot details.
Low_Stock_Report	Consumables where quantity < minimum_stock.
Calibration_Due_Report	Equipment whose calibration next_due_date is on or before the current date.

7. NORMALIZATION AND FUNCTIONAL DEPENDENCIES

Normalization is used to reduce redundancy and update anomalies. The design separates descriptive information from transactional information so that each relation has a clear purpose.

7.1 First Normal Form (1NF)

All attributes are atomic and there are no repeating groups inside a record. For example, one Booking record contains one user, one equipment item, one slot, one date and one status.

7.2 Second Normal Form (2NF)

The principal tables use single-attribute primary keys. Therefore, non-key attributes depend on the complete primary key rather than on part of a composite key. Entity details are separated into their own relations.

7.3 Third Normal Form (3NF)

Non-key descriptive attributes are stored with the entity they describe. Laboratory name and location are kept in Laboratory; category information is kept in Equipment_Category; vendor details are kept in Vendors; user details are kept in Users.

7.4 Functional Dependencies

DETERMINANT	FUNCTIONALLY DETERMINED ATTRIBUTES
lab_id	lab_name, location
category_id	category_name, description
equipment_id	equipment_name, category_id, lab_id, status
user_id	name, email, role
slot_id	start_time, end_time
booking_id	user_id, equipment_id, slot_id, booking_date, status
log_id	booking_id, equipment_id, user_id, actual_start, actual_end, purpose
consumable_id	lab_id, name, quantity, minimum_stock
calibration_id	equipment_id, calibration_date, next_due_date, status
breakdown_id	equipment_id, reported_by, reported_date, severity, description, status
vendor_id	vendor_name, contact, email
job_id	breakdown_id, vendor_id, equipment_id, start_date, completion_date, status, remarks

7.5 Normalization Summary

The separation of laboratories, equipment categories, equipment, users, time slots and vendors from booking and maintenance transactions reduces repeated data and supports consistent updates. The design therefore follows the intended 3NF structure used for the project documentation.

8. DATA DICTIONARY — TABLE STRUCTURE, DATA TYPES AND CONSTRAINTS

8.1 LABORATORY

COLUMN	DATA TYPE	KEY / CONSTRAINT
lab_id	INT	PK, NOT NULL
lab_name	VARCHAR(100)	NOT NULL
location	VARCHAR(100)	NULL

8.2 EQUIPMENT_CATEGORY

COLUMN	DATA TYPE	KEY / CONSTRAINT
category_id	INT	PK, NOT NULL
category_name	VARCHAR(100)	NOT NULL, UNIQUE
description	VARCHAR(255)	NULL

8.3 EQUIPMENT

COLUMN	DATA TYPE	KEY / CONSTRAINT
equipment_id	INT	PK, NOT NULL
equipment_name	VARCHAR(100)	NOT NULL
category_id	INT	FK, NOT NULL
lab_id	INT	FK, NOT NULL
status	VARCHAR(20)	NOT NULL, DEFAULT Active; CHECK

8.4 USERS

COLUMN	DATA TYPE	KEY / CONSTRAINT
user_id	INT	PK, NOT NULL
name	VARCHAR(100)	NOT NULL
email	VARCHAR(150)	NOT NULL, UNIQUE
role	VARCHAR(20)	NOT NULL

8.5 TIME_SLOTS

COLUMN	DATA TYPE	KEY / CONSTRAINT
slot_id	INT	PK, NOT NULL
start_time	TIME	NOT NULL
end_time	TIME	NOT NULL; CHECK start_time < end_time

8.6 BOOKINGS

COLUMN	DATA TYPE	KEY / CONSTRAINT
booking_id	INT	PK, NOT NULL
user_id	INT	FK, NOT NULL
equipment_id	INT	FK, NOT NULL
slot_id	INT	FK, NOT NULL
booking_date	DATE	NOT NULL
status	VARCHAR(20)	NOT NULL, DEFAULT Pending; CHECK

8.7 USAGE_LOGS

COLUMN	DATA TYPE	KEY / CONSTRAINT
log_id	INT	PK, NOT NULL
booking_id	INT	FK, NOT NULL
equipment_id	INT	FK, NOT NULL
user_id	INT	FK, NOT NULL
actual_start	DATETIME	NOT NULL
actual_end	DATETIME	NOT NULL
purpose	VARCHAR(255)	NULL

8.8 CONSUMABLES

COLUMN	DATA TYPE	KEY / CONSTRAINT
consumable_id	INT	PK, NOT NULL
lab_id	INT	FK, NOT NULL
name	VARCHAR(100)	NOT NULL
quantity	INT	NOT NULL; CHECK >= 0
minimum_stock	INT	NOT NULL; CHECK >= 0

8.9 CALIBRATIONS

COLUMN	DATA TYPE	KEY / CONSTRAINT
calibration_id	INT	PK, NOT NULL
equipment_id	INT	FK, NOT NULL
calibration_date	DATE	NOT NULL
next_due_date	DATE	NOT NULL; CHECK next_due_date > calibration_date
status	VARCHAR(20)	NOT NULL

8.10 BREAKDOWNS

COLUMN	DATA TYPE	KEY / CONSTRAINT
breakdown_id	INT	PK, NOT NULL
equipment_id	INT	FK, NOT NULL
reported_by	INT	FK, NOT NULL
reported_date	DATE	NOT NULL
severity	VARCHAR(20)	NOT NULL
description	VARCHAR(255)	NOT NULL
status	VARCHAR(20)	NOT NULL, DEFAULT Open

8.11 VENDORS

COLUMN	DATA TYPE	KEY / CONSTRAINT
vendor_id	INT	PK, NOT NULL
vendor_name	VARCHAR(100)	NOT NULL
contact	VARCHAR(20)	NULL
email	VARCHAR(150)	NULL

8.12 MAINTENANCE_JOBS

COLUMN	DATA TYPE	KEY / CONSTRAINT
job_id	INT	PK, NOT NULL
breakdown_id	INT	FK, NOT NULL
vendor_id	INT	FK, NOT NULL
equipment_id	INT	FK, NOT NULL
start_date	DATE	NOT NULL
completion_date	DATE	NULL; CHECK >= start_date
status	VARCHAR(20)	NOT NULL, DEFAULT Pending
remarks	VARCHAR(255)	NULL

9. SQL COMMANDS USED — DDL, DML AND DATABASE VALIDATION

9.1 Database Selection

```
CREATE DATABASE lab_equipment;  
USE lab_equipment;
```

9.2 Representative Table Creation

```
CREATE TABLE Equipment (  
    equipment_id INT PRIMARY KEY,  
    equipment_name VARCHAR(100) NOT NULL,  
    category_id INT NOT NULL,  
    lab_id INT NOT NULL,  
    status VARCHAR(20) NOT NULL DEFAULT 'Active',  
    FOREIGN KEY (category_id) REFERENCES Equipment_Category(category_id),  
    FOREIGN KEY (lab_id) REFERENCES Laboratory(lab_id)  
);
```

9.3 Constraint Definitions

```
ALTER TABLE Time_Slots  
ADD CONSTRAINT chk_time_slot  
CHECK (start_time < end_time);  
  
ALTER TABLE Bookings  
ADD CONSTRAINT chk_booking_status  
CHECK (status IN ('Pending', 'Approved', 'Rejected', 'Cancelled'));  
  
ALTER TABLE Equipment
```

```
ADD CONSTRAINT chk_equipment_status
CHECK (status IN ('Active', 'Maintenance', 'Broken', 'Retired'));
```

Additional CHECK constraints validate calibration dates, consumable quantities/minimum stock and maintenance completion dates.

9.4 Sample DML

```
INSERT INTO Equipment
(equipment_id, equipment_name, category_id, lab_id, status)
VALUES (106, 'Centrifuge C200', 5, 4, 'Active');
```

9.5 Views

```
CREATE VIEW Equipment_Overview AS
SELECT e.equipment_id, e.equipment_name, ec.category_name, l.lab_name, e.status
FROM Equipment e
JOIN Equipment_Category ec ON e.category_id=ec.category_id
JOIN Laboratory l ON e.lab_id=l.lab_id;
```

Four implemented views are Equipment_Overview, Booking_Report, Low_Stock_Report and Calibration_Due_Report.

9.6 Triggers

```
check_equipment_active
BEFORE INSERT ON Bookings

prevent_booking_overlap
BEFORE INSERT ON Bookings
```

The first trigger rejects booking of non-active equipment. The second checks for overlapping approved bookings for the same equipment and date.

10. QUERIES AND REPORTS

10.1 Active Equipment

```
SELECT * FROM Equipment WHERE status = 'Active';
```

10.2 Equipment with Category and Laboratory

```
SELECT e.equipment_id,e.equipment_name,ec.category_name,l.lab_name,e.status
FROM Equipment e
JOIN Equipment_Category ec ON e.category_id=ec.category_id
JOIN Laboratory l ON e.lab_id=l.lab_id;
```

10.3 Complete Booking Report

```
SELECT b.booking_id,u.name AS user_name,e.equipment_name,l.lab_name,
b.booking_date,ts.start_time,ts.end_time,b.status
FROM Bookings b
JOIN Users u ON b.user_id=u.user_id
JOIN Equipment e ON b.equipment_id=e.equipment_id
JOIN Laboratory l ON e.lab_id=l.lab_id
JOIN Time_Slots ts ON b.slot_id=ts.slot_id;
```

10.4 Most-Used Equipment

```
SELECT e.equipment_name,COUNT(ul.log_id) AS usage_count
FROM Equipment e JOIN Usage_Logs ul ON e.equipment_id=ul.equipment_id
GROUP BY e.equipment_id,e.equipment_name
ORDER BY usage_count DESC;
```

10.5 Low-Stock Consumables

```
SELECT consumable_id,name,quantity,minimum_stock
FROM Consumables WHERE quantity < minimum_stock;
```

10.6 Overdue Calibration

```
SELECT e.equipment_name,c.calibration_date,c.next_due_date
FROM Calibrations c JOIN Equipment e ON c.equipment_id=e.equipment_id
WHERE c.next_due_date < CURRENT_DATE;
```

10.7 Open Breakdowns

```
SELECT * FROM Breakdowns WHERE status='Open';
```

10.8 Booking Count per User

```
SELECT u.user_id,u.name,COUNT(b.booking_id) AS total_bookings
FROM Users u JOIN Bookings b ON u.user_id=b.user_id
GROUP BY u.user_id,u.name;
```

10.9 Booking Count per Equipment

```
SELECT e.equipment_id,e.equipment_name,COUNT(b.booking_id) AS total_bookings
FROM Equipment e LEFT JOIN Bookings b ON e.equipment_id=b.equipment_id
GROUP BY e.equipment_id,e.equipment_name;
```

10.10 Equipment Never Booked

```
SELECT equipment_id,equipment_name
FROM Equipment
WHERE equipment_id NOT IN (SELECT equipment_id FROM Bookings);
```

10.11 Users with Approved Bookings

```
SELECT user_id,name FROM Users
WHERE user_id IN (SELECT user_id FROM Bookings WHERE status='Approved');
```

10.12 Equipment with at Least Two Usage Sessions

```
SELECT e.equipment_id,e.equipment_name,COUNT(ul.log_id) AS usage_count
FROM Equipment e JOIN Usage_Logs ul ON e.equipment_id=ul.equipment_id
GROUP BY e.equipment_id,e.equipment_name
HAVING COUNT(ul.log_id)>=2;
```

10.13 Equipment with More Bookings than Average

```
SELECT e.equipment_id,e.equipment_name,COUNT(b.booking_id) AS total_bookings
FROM Equipment e JOIN Bookings b ON e.equipment_id=b.equipment_id
GROUP BY e.equipment_id,e.equipment_name
HAVING COUNT(b.booking_id) > (
SELECT AVG(booking_count) FROM (
SELECT COUNT(*) AS booking_count FROM Bookings GROUP BY equipment_id
) AS booking_summary
);
```

10.14 Breakdown Report

```
SELECT b.breakdown_id,e.equipment_name,u.name AS reported_by,  
b.reported_date,b.severity,b.description,b.status  
FROM Breakdowns b JOIN Equipment e ON b.equipment_id=e.equipment_id  
JOIN Users u ON b.reported_by=u.user_id;
```

10.15 Maintenance Report

```
SELECT mj.job_id,e.equipment_name,v.vendor_name,  
mj.start_date,mj.completion_date,mj.status  
FROM Maintenance_Jobs mj JOIN Equipment e ON mj.equipment_id=e.equipment_id  
JOIN Vendors v ON mj.vendor_id=v.vendor_id;
```

10.16 Presentation-II Query

The report includes the project query set used during database implementation. If the faculty-provided Presentation-II question differs from these project queries, its exact wording and output should be inserted in this section rather than inventing a question.

11. UI DESIGN AND SCREENSHOTS

The final project uses a web-based user interface as the front end. The UI presents database information through dashboard and table-based screens and supports the project demonstration of CRUD operations.

11.1 UI Modules

SCREEN / MODULE	PURPOSE
Dashboard	Summary of laboratory resources, bookings and maintenance information.
Equipment	View and manage equipment records.
Bookings	View booking records, dates, slots and booking status.
Consumables	View quantities and minimum-stock thresholds.
Maintenance	View breakdown and maintenance-job information.
Reports	Present database query results and operational reports.

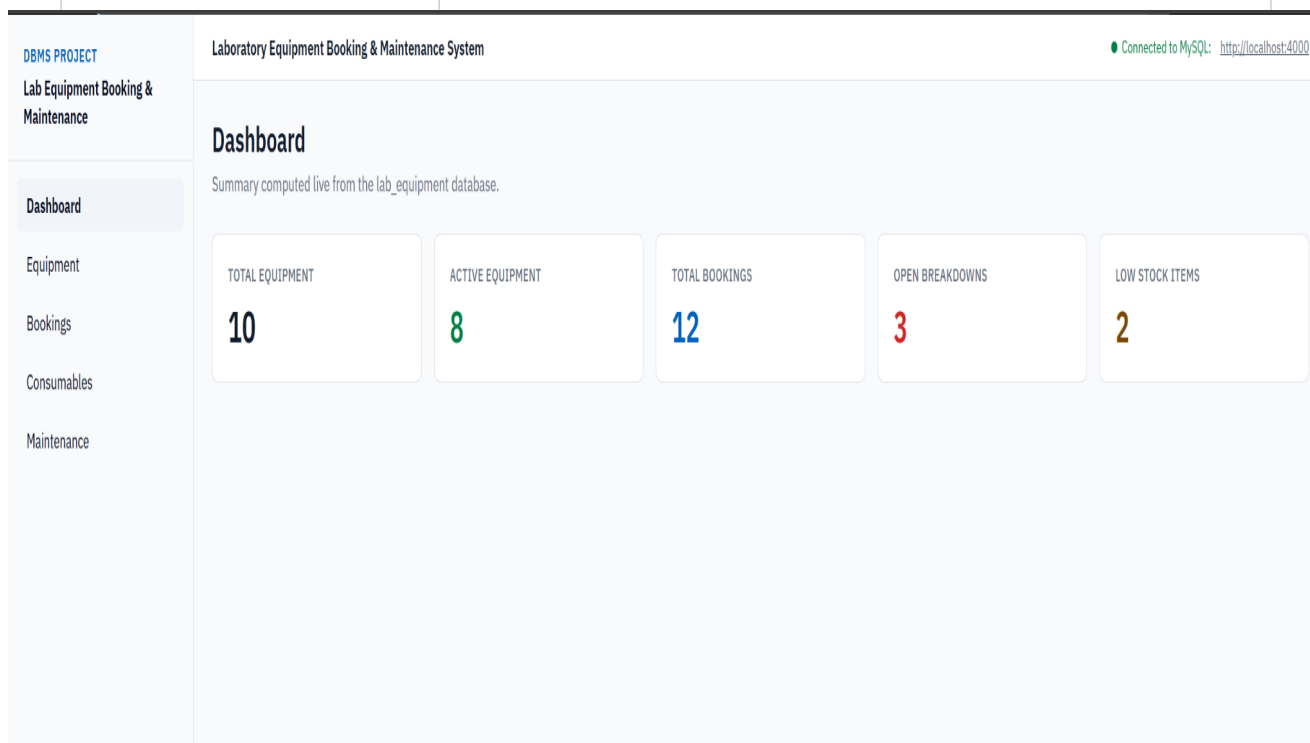


Figure 2. Project dashboard / database summary screen.

DBMS PROJECT

Lab Equipment Booking & Maintenance

Dashboard

Equipment

Bookings

Consumables

Maintenance

Laboratory Equipment Booking & Maintenance System

Connected to MySQL: http://localhost:4000

Equipment

Equipment > Equipment_Category > Laboratory

+ Add Equipment

EQUIPMENT ID	EQUIPMENT NAME	CATEGORY	LABORATORY	STATUS	ACTIONS
101	DSO-1000	Oscilloscope	Physics Lab	Active	Delete
102	DSO-2000	Oscilloscope	Physics Lab	Active	Delete
103	Digital Multimeter	Multimeter	Electronics Lab	Active	Delete
104	Microscope X1	Microscope	Electronics Lab	Maintenance	Delete
105	3D Printer Pro	3D Printer	Computer Lab	Active	Delete
106	Centrifuge C200	Centrifuge	Chemistry Lab	Active	Delete
107	SolderPro X2	Soldering Station	Electronics Lab	Active	Delete
108	Function Generator FG1	Function Generator	Physics Lab	Active	Delete
109	Centrifuge C300	Centrifuge	Chemistry Lab	Broken	Delete
110	solderpero x3	Soldering Station	Electronics Lab	Active	Delete

Figure 3. Equipment management screen.

DBMS PROJECT

Lab Equipment Booking & Maintenance

Dashboard

Equipment

Bookings

Consumables

Maintenance

Laboratory Equipment Booking & Maintenance System

● Connected to MySQL: http://localhost:4000

Bookings

Bookings > Users > Equipment > Laboratory > Time_Slots

+ New Booking

BOOKING ID	USER	EQUIPMENT	LABORATORY	DATE	START	END	STATUS	ACTIONS
1001	Rahul	DSO-1000	Physics Lab	2026-08-27	10:00	11:00	Approved	Delete
1002	Priya	Digital Multimeter	Electronics Lab	2026-08-27	11:00	12:00	Pending	Delete
1003	Rahul	DSO-2000	Physics Lab	2026-08-28	09:00	10:00	Approved	Delete
1004	Priya	DSO-1000	Physics Lab	2026-08-29	10:00	11:00	Approved	Delete
1005	Priya	DSO-1000	Physics Lab	2026-08-27	10:00	11:00	Pending	Delete
1006	Ananya	Centrifuge C200	Chemistry Lab	2026-08-31	09:00	10:00	Approved	Delete
1007	Vikram	SolderPro X2	Electronics Lab	2026-08-31	10:00	11:00	Approved	Delete
1008	Neha	Function Generator FG1	Physics Lab	2026-08-31	11:00	12:00	Approved	Delete
1009	Ananya	DSO-1000	Physics Lab	2026-09-01	14:00	15:00	Pending	Delete
1010	Vikram	Digital Multimeter	Electronics Lab	2026-09-01	15:00	16:00	Approved	Delete
1011	Meera	Centrifuge C200	Chemistry Lab	2026-09-02	11:00	12:00	Approved	Delete
1012	Rohit	Function Generator FG1	Physics Lab	2026-09-02	14:00	15:00	Rejected	Delete

Figure 4. Booking management and booking-status screen.

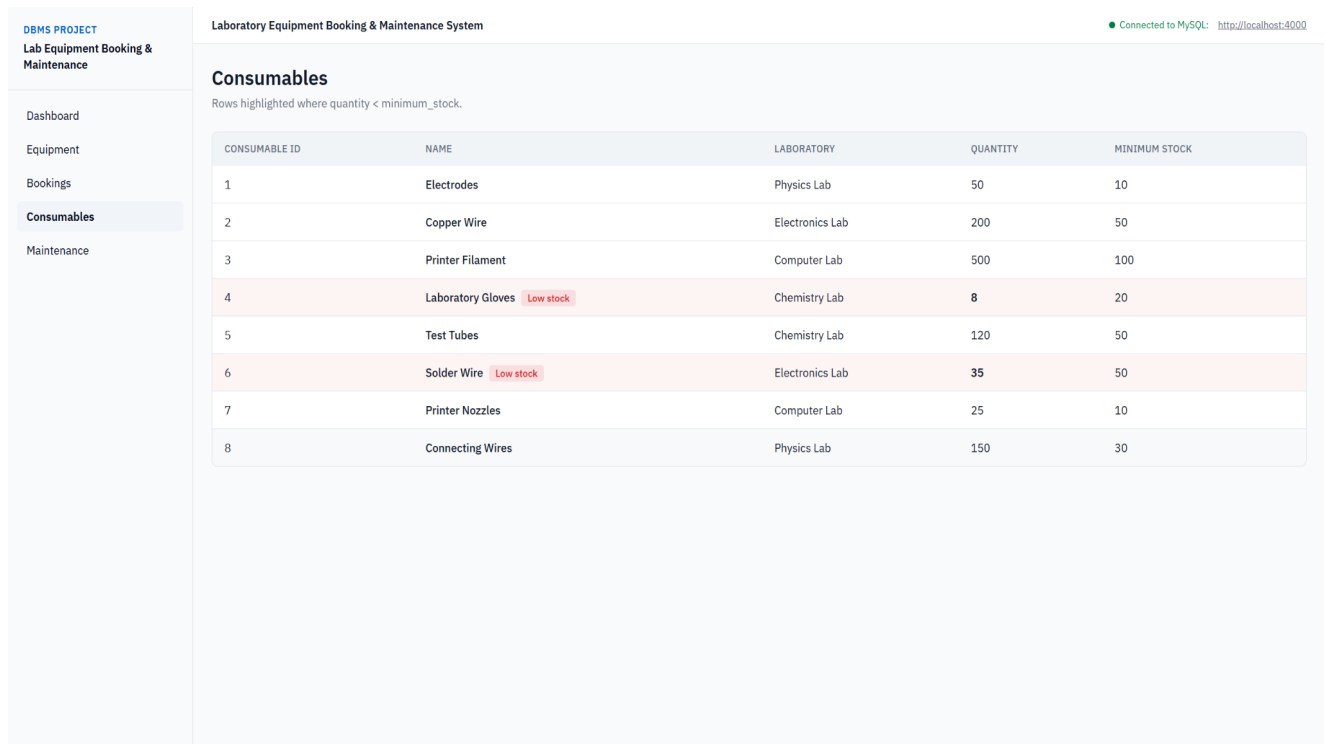


Figure 5. Consumables / inventory screen.

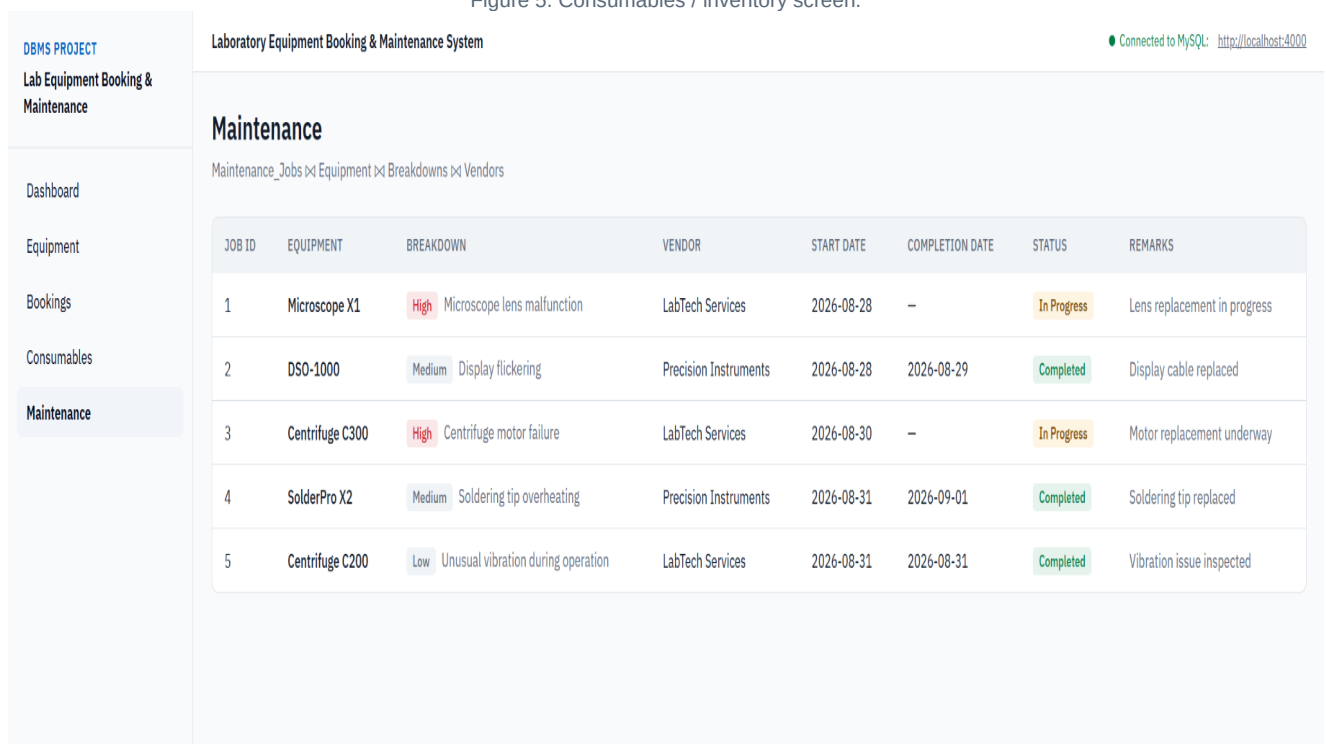


Figure 6. Maintenance and related equipment information screen.

11.2 Required Before/After Evidence

- Equipment INSERT — before and after
- Equipment VIEW — inserted row visible in UI
- MySQL SELECT — inserted row visible in database
- Equipment DELETE — before and after
- Booking record visible after creation
- Report output visible after running query
- Screenshots of each final UI screen/module for project submission

12. IMPLEMENTATION DETAILS

12.1 Technology Stack

LAYER	TECHNOLOGY	ROLE
UI	Web-based user interface	Forms, tables, dashboard and CRUD interaction
Database	MySQL 8.0.46	Persistent relational storage
Database Administration	MySQL Workbench / MySQL CLI	SQL execution, verification and schema work
Development	Project source files / IDE	Application development
Version Control	Git / GitHub	Project submission

12.2 Database Connectivity

The application communicates with the MySQL database and displays records from the lab_equipment database. The database can be verified independently using MySQL CLI and MySQL Workbench.

12.3 CRUD and Transaction Handling

- INSERT operations add new records to the relevant base table.
- SELECT operations retrieve and report database records.
- UPDATE operations can modify existing records where supported by the UI.
- DELETE operations remove test records while respecting foreign-key relationships.
- Database validation rules reject invalid values before they are stored.

12.4 Database Integrity Rules

- Primary keys uniquely identify records.
- Foreign keys enforce valid relationships between tables.
- UNIQUE constraints prevent duplicate category names and user emails.
- NOT NULL constraints require essential values.
- CHECK constraints restrict invalid status, time, date and quantity values.
- Triggers enforce booking-specific business rules.

13. TESTING — TEST CASES AND RESULTS

ID	TEST	EXPECTED RESULT	STATUS
TC-01	Database connection / schema access	Database opens and tables are accessible	PASS
TC-02	Valid equipment insertion	New equipment row created	PASS
TC-03	Equipment viewing	Inserted/existing row visible	PASS
TC-04	Equipment deletion test	Test row removed from database	PASS / verify
TC-05	Foreign-key integrity	Invalid referenced ID rejected	PASS
TC-06	Invalid time slot	CHECK rejects start_time >= end_time	PASS
TC-07	Invalid booking status	CHECK rejects unsupported status	PASS
TC-08	Broken equipment booking	Trigger rejects non-active equipment	PASS
TC-09	Overlapping approved booking	Trigger rejects overlap	PASS
TC-10	Negative consumable quantity	CHECK rejects negative quantity	PASS
TC-11	Invalid calibration date	CHECK rejects invalid due date	PASS
TC-12	Invalid maintenance date	CHECK rejects completion before start	PASS
TC-13	Low-stock report	Rows below minimum returned	PASS
TC-14	Calibration report	Due/overdue equipment returned	PASS
TC-15	Breakdown report	Joined breakdown data returned	PASS
TC-16	Maintenance report	Joined maintenance data returned	PASS

13.1 Testing Approach

Testing is performed at database and UI/database levels. Valid records are inserted and viewed, while invalid values are deliberately tested against CHECK constraints and triggers. Reporting queries and views are also executed to confirm that related data can be retrieved correctly.

14. CONCLUSION AND FUTURE ENHANCEMENTS

14.1 Conclusion

The Laboratory Equipment Booking and Maintenance System demonstrates the design and implementation of a normalized relational database for managing shared laboratory resources. The project connects 12 principal tables through primary and foreign keys and applies integrity constraints to maintain consistent data.

The system supports equipment booking, usage logging, consumable monitoring, calibration tracking, breakdown reporting, maintenance jobs and vendor management. SQL queries, views and triggers provide reporting and validation capabilities. The web-based UI provides a practical front end for demonstrating the database.

14.2 Future Enhancements

- Role-based login and access control
- Calendar-based equipment scheduling
- Automatic calibration and maintenance reminders
- Email / notification integration
- Advanced equipment utilization analytics
- Automated inventory alerts
- Cloud deployment and remote database access
- Audit logging and automated backups
- Integration with institutional laboratory systems

15. REFERENCES

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4. DBMS Course Project and Presentation requirements, Woxsen University.
5. Uploaded reference DBMS project report used for report organization and presentation style.

APPENDIX A — COMPLETE POPULATED DATABASE TABLES

This appendix includes the populated records of all 12 base tables in the implemented project database. The reference report also places populated database tables in an appendix; this expanded version follows that structure while including every project table.

A.1 LABORATORY — POPULATED RECORDS

lab_id	lab_name	location
1	Physics Lab	Block A
2	Electronics Lab	Block B
3	Computer Lab	Block C
4	Chemistry Lab	Block D
5	Mechanical Lab	Block E

A.2 EQUIPMENT_CATEGORY — POPULATED RECORDS

category_id	category_name	description
1	Oscilloscope	Digital oscilloscopes for waveform measurement
2	Microscope	Laboratory microscopes
3	Multimeter	Digital electrical measurement equipment
4	3D Printer	Additive manufacturing equipment
5	Centrifuge	Laboratory centrifuge equipment
6	Soldering Station	Electronic soldering equipment
7	Function Generator	Signal generation equipment

A.3 EQUIPMENT — POPULATED RECORDS

equipment_id	equipment_name	category_id	lab_id	status
101	DSO-1000	1	1	Active
102	DSO-2000	1	1	Active
103	Digital Multimeter	3	2	Active
104	Microscope X1	2	1	Maintenance
105	3D Printer Pro	4	3	Active
106	Centrifuge C200	5	4	Active
107	SolderPro X2	6	2	Active
108	Function Generator FG1	7	2	Active
109	Centrifuge C300	5	4	Broken
110	SolderPro X3	6	2	Retired

A.4 USERS — POPULATED RECORDS

user_id	name	email	role
1	Rahul	rahul@example.com	Student
2	Priya	priya@example.com	Faculty
3	Arjun	arjun@example.com	Technician
4	Admin	admin@example.com	Administrator
5	Ananya	ananya@example.com	Student
6	Vikram	vikram@example.com	Student
7	Neha	neha@example.com	Faculty
8	Karan	karan@example.com	Technician
9	Meera	meera@example.com	Student
10	Rohit	rohit@example.com	Faculty

A.5 TIME_SLOTS — POPULATED RECORDS

slot_id	start_time	end_time
1	09:00	10:00
2	10:00	11:00
3	11:00	12:00
4	14:00	15:00
5	15:00	16:00
6	16:00	17:00
7	17:00	18:00
8	18:00	19:00

A.6 BOOKINGS — POPULATED RECORDS

booking_id	user_id	equipment_id	slot_id	booking_date	status
1001	1	101	2	2026-08-28	Approved
1002	2	103	3	2026-08-28	Pending
1003	1	102	1	2026-08-29	Approved
1004	2	101	2	2026-08-30	Approved
1005	2	101	2	2026-08-28	Pending
1006	5	106	1	2026-09-01	Approved
1007	6	107	2	2026-09-01	Approved
1008	7	108	3	2026-09-01	Approved
1009	5	101	4	2026-09-02	Pending
1010	6	103	5	2026-09-02	Approved
1011	9	106	3	2026-09-03	Approved
1012	10	108	4	2026-09-03	Rejected

A.7 USAGE_LOGS — POPULATED RECORDS

log_id	booking_id	equipment_id	user_id	actual_start	actual_end	purpose
1	1001	101	1	2026-08-28 10:05	2026-08-28 10:52	Physics experiment
2	1006	106	5	2026-09-01 09:05	2026-09-01 09:55	Centrifuge sample preparation
3	1007	107	6	2026-09-01 10:04	2026-09-01 10:48	Circuit assembly
4	1008	108	7	2026-09-01 11:03	2026-09-01 11:51	Signal testing
5	1010	103	6	2026-09-02 15:02	2026-09-02 15:44	Electrical measurement
6	1011	106	9	2026-09-03 11:04	2026-09-03 11:49	Sample separation

A.8 CONSUMABLES — POPULATED RECORDS

consumable_id	lab_id	name	quantity	minimum_stock
1	1	Electrodes	50	10
2	2	Copper Wire	200	50
3	3	Printer Filament	500	100
4	4	Laboratory Gloves	8	20
5	4	Test Tubes	120	50
6	2	Solder Wire	35	50
7	3	Printer Nozzles	25	10
8	1	Connecting Wires	150	30

A.9 CALIBRATIONS — POPULATED RECORDS

calibration_id	equipment_id	calibration_date	next_due_date	status
1	101	2026-01-15	2027-01-15	Valid
2	103	2026-03-10	2027-03-10	Valid
3	104	2025-08-01	2026-08-01	Overdue
4	102	2026-02-10	2027-02-10	Valid
5	105	2026-04-20	2027-04-20	Valid
6	106	2026-05-15	2027-05-15	Valid
7	107	2025-09-01	2026-09-01	Due
8	108	2026-01-20	2027-01-20	Valid

A.10 BREAKDOWNS — POPULATED RECORDS

breakdown_id	equipment_id	reported_by	reported_date	severity	description	status
1	104	1	2026-08-28	High	Microscope lens malfunction	Open
2	101	2	2026-08-29	Medium	Display flickering	Resolved
3	109	5	2026-08-30	High	Centrifuge motor failure	Open
4	107	6	2026-08-31	Medium	Soldering tip overheating	Open
5	106	8	2026-09-01	Low	Unusual vibration during operation	Resolved

A.11 VENDORS — POPULATED RECORDS

vendor_id	vendor_name	contact	email
1	LabTech Services	9876543210	service@labtech.com
2	Precision Instruments	9876501234	support@precision.com

A.12 MAINTENANCE_JOBS — POPULATED RECORDS

job_id	breakdown_id	vendor_id	equipment_id	start_date	completion_date	status	remarks
1	1	1	104	2026-08-29	—	In Progress	Lens replacement in progress
2	2	2	101	2026-08-29	2026-08-30	Completed	Display cable replaced
3	3	1	109	2026-08-31	—	In Progress	Motor replacement underway
4	4	2	107	2026-09-01	2026-09-02	Completed	Soldering tip replaced
5	5	1	106	2026-09-01	2026-09-01	Completed	Vibration issue inspected

APPENDIX B — DATABASE POPULATION SUMMARY

TABLE	TOTAL RECORDS	IDENTIFICATION / RANGE
Laboratory	5	1–5
Equipment_Category	7	1–7
Equipment	10	101–110
Users	10	1–10
Time_Slots	8	1–8
Bookings	12	1001–1012
Usage_Logs	6	1–6
Consumables	8	1–8
Calibrations	8	1–8
Breakdowns	5	1–5
Vendors	2	1–2
Maintenance_Jobs	5	1–5

Base tables: 12. Total populated base-table records in this project: 86.

APPENDIX C — VIEWS, TRIGGERS AND CONSTRAINTS

C.1 Views

VIEW	LOGIC / PURPOSE
Equipment_Overview	Joins Equipment, Equipment_Category and Laboratory.
Booking_Report	Joins Bookings, Users, Equipment, Laboratory and Time_Slots.
Low_Stock_Report	Returns consumables where quantity is below minimum_stock.
Calibration_Due_Report	Returns equipment whose next calibration date is due or overdue.

C.2 Triggers

TRIGGER	EVENT	PURPOSE
check_equipment_active	BEFORE INSERT ON Bookings	Rejects booking when referenced equipment is not Active.
prevent_booking_overlap	BEFORE INSERT ON Bookings	Rejects an Approved booking that overlaps an existing Approved booking for the same equipment/date.

C.3 CHECK Constraints

CONSTRAINT	TABLE	RULE
chk_time_slot	Time_Slots	start_time < end_time
chk_booking_status	Bookings	status is Pending, Approved, Rejected or Cancelled
chk_equipment_status	Equipment	status is Active, Maintenance, Broken or Retired
chk_calibration_dates	Calibrations	next_due_date > calibration_date
chk_consumable_quantity	Consumables	quantity >= 0
chk_consumable_minimum_stock	Consumables	minimum_stock >= 0
chk_maintenance_dates	Maintenance_Jobs	completion_date is NULL or completion_date >= start_date

APPENDIX D — SUBMISSION CHECKLIST

- Final DBMS project report PDF
- ER diagram image
- Presentation-I PDF/PPT
- Presentation-II PDF/PPT
- Presentation-III PDF/PPT
- Complete SQL script
- Database data / sample records
- Web-based UI source code
- UI screenshots
- Insert / view / delete evidence
- GitHub repository link / README